

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.3	10.3	1.5	0.5	0.2	56.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.6	9	0.6	0.4	0.2	56.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.3	10.5	1.6	0.4	0.2	57
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.6	9.3	0.5	0.1	0.2	56.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.7	10.2	1.3	0.5	0.2	56.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.6	8.1	0.6	0.2	0.2	56.3
U2 - Efficacy additive alpha prior Beta(1, 1)	30.6	10.5	1.9	0.3	0.2	56.5
U3 - Efficacy additive alpha prior Beta(1, 1)	34.6	8.2	0.5	0.2	0.2	56.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.4	3.9	2.1	0.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.3	3.9	2.1	0.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.3	3.9	2.2	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.2	3.9	2.2	0.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.3	3.9	2.1	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.4	3.9	2.2	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.3	3.9	2.2	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.4	3.9	2.2	0.9	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.1	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	2	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	2	0.6	0.4	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.9	16	31.9	11.6	3.8	1.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	38.3	16.1	30.8	10.1	2.9	1.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.6	15.4	33.2	11	4	1.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	38.8	14.8	31.7	9.8	3	1.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.5	14.4	33.8	11.3	4.1	1.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	40.1	13.6	32	9.3	3.2	1.8
U2 - Efficacy additive alpha prior Beta(1, 1)	34.8	14.3	34	10.9	4.2	1.8
U3 - Efficacy additive alpha prior Beta(1, 1)	40.4	13.9	31.4	8.7	3.8	1.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	5.1	7.2	3.9	2.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	5.1	7.2	3.9	2.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	5	7.3	3.8	2.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	5	7.3	3.9	2.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.4	4.9	7.3	3.9	2.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	4.9	7.2	3.9	2.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.5	4.8	7.3	3.9	2.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.4	4.9	7.3	3.8	2.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.4	1.3	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.4	1.3	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.4	1.3	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.4	1.3	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.3	1.4	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.3	1.3	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.3	1.3	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.3	1.3	0.9	0.3	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	40.4	21.1	8.2	2.8	10
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.2	41.4	20.9	5.4	2.1	10
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.5	40.1	21.7	7.4	3.3	10
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.8	40.7	20	6.3	2.2	10
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.3	36.1	23.8	8.4	3.3	10.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22	38.3	21	6.6	2.2	9.9
U2 - Efficacy additive alpha prior Beta(1, 1)	17.9	37.3	23.5	7.8	3.6	9.9
U3 - Efficacy additive alpha prior Beta(1, 1)	21.7	39.7	20.4	6.2	2	10
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	7.5	6.1	3.6	2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	7.4	6.2	3.6	2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	7.3	6.2	3.6	2.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	7.3	6.3	3.6	2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	7.1	6.3	3.6	2.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	7.1	6.4	3.6	2.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.7	7.1	6.3	3.6	2.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.7	7.2	6.3	3.5	2.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	1.3	1	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	1.3	1	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	1.3	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.2	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.3	1	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.3	1	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.3	1	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.3	1	0.4	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.8	7.4	21.4	42.6	19.9	1.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.5	9	19.1	40.8	17.7	1.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.8	6.9	21.3	41.8	21.3	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.5	7.8	18.5	41.6	18.7	1.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.3	7.5	20.4	42.9	21	1.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.4	8.4	18.7	41.3	18.3	1.9
U2 - Efficacy additive alpha prior Beta(1, 1)	6.2	6.9	20.4	43.3	21.3	1.9
U3 - Efficacy additive alpha prior Beta(1, 1)	11.1	8.7	19	40.8	18.5	1.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	4.2	6.5	8	6.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	4.2	6.5	7.9	6.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	4.2	6.5	7.8	6.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	4.2	6.5	7.8	6.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	4.2	6.5	7.8	6.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	4.3	6.4	7.9	6.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.5	4.2	6.6	7.7	6.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.5	4.2	6.5	7.8	6.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.4	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.4	2.2	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.4	2.1	0.7	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	35.3	26.3	18.1	11.2	4.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	41.1	23.4	16.3	7.2	4.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	32.7	27.1	18	12.1	4.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	40.8	24.4	14.2	6.8	4.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	32.5	26.9	18.2	12.7	4.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	39.4	25.1	14.6	7.4	4.2
U2 - Efficacy additive alpha prior Beta(1, 1)	5.9	31.1	27.5	18.9	12.4	4.2
U3 - Efficacy additive alpha prior Beta(1, 1)	9.4	39	25.5	14.8	7.1	4.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	6.4	6.7	5.7	5.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	6.4	6.7	5.7	5.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	6.2	6.7	5.7	5.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	6.3	6.7	5.7	5.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	6.2	6.6	5.6	5.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	6.2	6.7	5.6	5.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.1	6.1	6.7	5.7	5.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	5	6.2	6.6	5.7	5.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.4	1.8	2.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.1	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.3	1.7	2.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.3	1.7	2.1	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.7	2.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.7	2.1	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	1.7	2	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	1.7	2.1	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	67.6	14.2	3.2	1.2	0.5	13.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	68.7	13.3	3.1	1	0.4	13.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.8	12.9	4	1.3	0.5	13.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.4	12.6	2.4	0.7	0.4	13.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.6	11.8	3.9	1.6	0.5	13.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	71.7	10.9	2.6	1	0.4	13.4
U2 - Efficacy additive alpha prior Beta(1, 1)	69.2	11.5	3.8	1.4	0.4	13.7
U3 - Efficacy additive alpha prior Beta(1, 1)	72	10.2	2.8	1.3	0.4	13.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.6	3.9	2.5	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.6	3.9	2.5	1.1	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.6	3.8	2.6	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.6	3.9	2.5	1.1	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.7	3.7	2.5	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.7	3.7	2.6	1.1	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.7	3.7	2.6	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.7	3.6	2.6	1.1	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.5	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.5	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.5	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.5	0.5	0.3	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.4	36.6	20.5	10.6	3.7	2.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.2	39	18.9	8.6	3	2.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.2	34.9	20.6	12.1	4	2.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.6	38.3	18.6	9.1	3.1	2.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.4	33.8	22	10.1	4.5	2.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30	36.1	20.3	8.2	3.2	2.2
U2 - Efficacy additive alpha prior Beta(1, 1)	26.5	34.6	21.9	10.7	4	2.3
U3 - Efficacy additive alpha prior Beta(1, 1)	29.5	36.7	19.3	9.1	3.2	2.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	7	5.6	3.8	2.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	6.9	5.7	3.7	2.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	6.8	5.7	3.8	2.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	6.8	5.6	3.7	2.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	6.7	5.7	3.7	2.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	6.7	5.8	3.6	2.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.9	6.6	5.8	3.7	2.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.9	6.6	5.8	3.7	2.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1.1	0.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1.1	0.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1.1	1	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1.1	1	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.5	1.1	0.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.5	1.1	0.9	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.5	1.1	0.9	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.5	1.1	1	0.4	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	43.2	9.2	2.8	1.2	0.3	43.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	45	8.7	2	0.9	0.2	43.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	43	9.4	2.8	1	0.4	43.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	45.4	7.7	2	1.2	0.2	43.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	42.5	9.9	2.5	1	0.4	43.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	45.8	7.5	1.7	1	0.3	43.7
U2 - Efficacy additive alpha prior Beta(1, 1)	43.6	9.2	2.4	1.2	0.3	43.3
U3 - Efficacy additive alpha prior Beta(1, 1)	46	7.3	2	1.1	0.2	43.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.3	3.7	1.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.4	3.7	1.9	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.3	3.7	1.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.3	3.7	1.9	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.5	3.6	1.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.5	3.6	1.8	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.5	3.6	1.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.5	3.6	1.9	0.8	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.6	0.4	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.6	0.4	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.6	0.4	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.7	0.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.6	0.4	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.7	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.4	0.2	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.4	10.5	17.5	13.2	26.8	18.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	10.6	14.8	12	23	18.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.1	10.1	18	13	27	18.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20	11.4	14.5	12	23.3	18.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.3	9.8	17.6	13.3	26.3	18.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.8	10.5	14.8	11.4	23.6	18.9
U2 - Efficacy additive alpha prior Beta(1, 1)	14.4	9.6	16.6	13.2	27.2	19
U3 - Efficacy additive alpha prior Beta(1, 1)	20	11.8	13.3	12.5	23.8	18.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	4.3	5.3	4	5.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	4.3	5.3	4	5.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	4.2	5.3	4	5.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	4.3	5.4	3.9	5.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	4.2	5.4	3.9	5.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	4.2	5.4	3.9	5.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.8	4.2	5.4	3.9	5.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.8	4.2	5.3	4	5.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.6	1.3	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.6	1.3	1.4	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.6	1.3	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.6	1.3	1.4	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	1.3	1.4	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.6	1.3	1.4	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.6	1.3	1.4	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.6	1.3	1.4	0.6	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	31.7	11.8	4.6	2.4	38.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	32.5	10.7	4.6	1.5	38.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.2	31.3	11.7	4.4	2.4	38
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.9	32.6	10.6	4.1	1.6	38.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	30.5	12.5	4.9	2.7	38.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.5	32.9	11.2	3.3	1.9	38.2
U2 - Efficacy additive alpha prior Beta(1, 1)	11.3	31.3	11.8	4.7	2.7	38.2
U3 - Efficacy additive alpha prior Beta(1, 1)	13.1	31.9	11	3.9	2.1	38
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.7	5.8	4.1	2.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.7	5.9	4	2.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	5.8	4.1	2.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	5.7	4.1	2.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.8	5.7	4.2	2.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.8	5.7	4.1	2.2	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.8	5.8	4.1	2.1	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.8	5.7	4.2	2.1	1.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.2	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.2	0.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.2	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.2	0.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.2	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.2	0.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.2	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.3	0.8	0.6	0.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	65.9	22	4.6	0.5	0	7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.5	20.6	2.4	0.7	0	6.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	66.5	21.4	4.6	0.6	0	6.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.4	19.3	2.8	0.7	0	6.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	68	20	4.5	0.6	0	6.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	72.3	17.2	3.1	0.6	0	6.8
U2 - Efficacy additive alpha prior Beta(1, 1)	68.5	20.2	3.9	0.5	0	6.9
U3 - Efficacy additive alpha prior Beta(1, 1)	72.1	17.3	3	0.6	0	7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.4	5.8	3.5	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.4	5.8	3.5	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.5	5.7	3.5	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.4	5.7	3.6	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.6	5.6	3.5	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.6	5.6	3.5	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.6	5.6	3.5	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.6	5.6	3.5	1.1	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.4	0.8	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.4	0.8	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.4	0.8	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.4	0.8	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.4	0.8	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.4	0.8	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.4	0.8	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.4	0.8	0.4	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.2	12.3	10	36	24.2	3.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.2	12.2	6.8	34.4	22.1	3.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.8	13.3	10.4	35.4	24.8	3.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.4	11.8	6.6	35.3	22.7	3.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.2	13.9	9.8	35.6	24.2	3.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21	12.5	6.4	33.5	23.3	3.3
U2 - Efficacy additive alpha prior Beta(1, 1)	12.6	13.1	9.9	35.9	25.1	3.4
U3 - Efficacy additive alpha prior Beta(1, 1)	21.3	12.7	6.4	32.9	23.4	3.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	4.5	5.3	7.3	7.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	4.6	5.2	7.3	7.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	4.5	5.2	7.2	7.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	4.5	5.2	7.4	7.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	4.6	5.2	7.2	7.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	4.5	5.2	7.3	7.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	5	4.6	5.2	7.2	7.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.1	4.5	5.2	7.3	7.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.9	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.9	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.9	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.9	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.9	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.9	1.3	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.9	1.4	2.2	0.8	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.8	32.8	13.4	4.6	2	33.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.8	32.7	12.6	4.4	1.6	32.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.5	31.3	14.6	5.1	2	33.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.9	32.5	12.8	3.8	1.6	33.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.3	31.9	15.1	4.3	2.2	33.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16	33	12.2	4.2	1.5	33.1
U2 - Efficacy additive alpha prior Beta(1, 1)	12.9	31.7	14.9	5.1	1.9	33.5
U3 - Efficacy additive alpha prior Beta(1, 1)	15.7	32.4	13	4	1.7	33.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	6.1	4.5	2.5	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	6.1	4.5	2.4	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.5	6	4.5	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	6.1	4.5	2.4	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	6	4.6	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.4	6.1	4.6	2.4	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.5	6	4.5	2.4	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.5	6	4.5	2.4	1.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.5	0.9	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.5	0.9	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.5	0.9	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.5	0.9	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.5	0.9	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	0.9	0.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	0.9	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	0.9	0.7	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.6	14.5	12.8	27.8	18	7.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.5	15.1	9	25.3	12.7	7.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.6	16.6	13.3	27.5	16.7	7.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.2	14.9	9	24.4	13.2	7.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.2	16.1	14.1	26.3	16.9	7.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.2	14.7	9.4	23.8	13.5	7.4
U2 - Efficacy additive alpha prior Beta(1, 1)	18.6	15.9	14.9	27	16.3	7.3
U3 - Efficacy additive alpha prior Beta(1, 1)	29.9	15.5	9.8	24.3	13.1	7.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.7	5.2	5.8	6.1	5.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.7	5.2	5.8	6.1	5.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.7	5.2	5.9	6	5.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.7	5.2	5.9	6.1	5.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.6	5.2	5.8	6	5.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.7	5.1	5.9	5.9	5.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.7	5.2	5.9	6	5.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.7	5.2	5.8	5.9	5.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.1	1.5	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.1	1.5	2	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.1	1.4	2	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.1	1.5	2	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.1	1.5	2	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.1	1.5	2	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.1	1.5	2	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.2	1.5	2	0.7	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	52.4	24.9	10.9	3.6	1.2	7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	63.8	20.5	5.7	2.4	0.4	7.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	53.3	24.5	10.7	3.5	1	7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	66.6	18.8	5.7	1.3	0.6	7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	52.8	24.5	10.7	3.9	1.1	7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	67	17.9	5.8	1.8	0.4	7.1
U2 - Efficacy additive alpha prior Beta(1, 1)	56	22.7	9.4	4	0.9	7
U3 - Efficacy additive alpha prior Beta(1, 1)	68.2	17.8	5.2	1.4	0.4	7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.6	6.6	5.9	3.3	1.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.7	6.6	5.8	3.3	1.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.5	6.7	5.9	3.3	1.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.5	6.6	5.9	3.4	1.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	6.6	5.9	3.3	1.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	6.6	6	3.3	1.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.6	6.6	5.9	3.3	1.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.7	6.6	5.9	3.3	1.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.4	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.4	1.6	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.4	1.5	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.4	1.6	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.4	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.5	1.6	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.5	1.6	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.5	1.6	1.3	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.4	23.2	16.2	16.6	12.5	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	38.2	25.7	15.6	12.9	7.5	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	31	22.6	17.4	16.4	12.5	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	38.8	25.9	15.4	12.5	7.3	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.8	22.3	18.1	15.1	12.6	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	38.6	25.4	17.1	12.3	6.5	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	32.4	20.9	18.3	15.8	12.5	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	38.7	26	17.9	11.1	6.2	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.1	5.4	5.4	5.8	7.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6	5.4	5.5	5.8	7.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.1	5.3	5.4	5.8	7.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6	5.4	5.4	5.8	7.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.1	5.3	5.3	5.7	7.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.1	5.3	5.3	5.7	7.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.2	5.2	5.3	5.7	7.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.1	5.2	5.4	5.7	7.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.1	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.1	1.8	2.6	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.1	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.1	1.8	2.6	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.1	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.1	1.8	2.6	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.1	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	3	1.8	2.6	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.3	49.6	17.9	1.9	0	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	34	49.1	14.8	1.8	0	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	33	47.5	17.3	1.9	0	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.5	47.3	15.4	1.4	0.1	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.7	46.2	18.6	2.1	0.1	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.2	45.8	16.2	1.5	0	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	32.9	45.8	19.2	1.8	0	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	37.6	44.4	16.2	1.4	0.1	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.2	9.7	7.3	2.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	9.7	7.2	2.3	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	9.4	7.4	2.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	9.4	7.5	2.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	9.3	7.6	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	9.3	7.5	2.3	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.5	9.3	7.5	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.4	9.2	7.5	2.3	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.8	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.8	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.8	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.7	1.1	0.6	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	7.5	9.2	14.2	58.5	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.1	10	9.6	11.8	51.7	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.1	9.8	8.4	13.5	58.4	0.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16	10.2	8.6	12.4	52	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.5	10	10	13.7	58	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16	9.4	8.9	12.7	52.2	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	7.8	9.9	10.2	12.7	58.6	0.8
U3 - Efficacy additive alpha prior Beta(1, 1)	16.3	10.6	8.6	12	51.7	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	3.8	4.6	5.5	12.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	3.9	4.6	5.5	12.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.7	3.8	4.5	5.5	12.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	3.8	4.6	5.5	12.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	3.8	4.5	5.4	12.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	3.8	4.5	5.4	12.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.8	3.8	4.5	5.4	12.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.7	3.8	4.5	5.4	12.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.3	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.3	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.3	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.3	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.4	2.3	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.6	51.9	17.5	1.6	0	8.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.3	52.2	15.1	1.2	0	8.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21	51.3	17.8	1.5	0	8.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.4	50.7	15.1	1.5	0	8.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.4	51.4	18.2	1.5	0.1	8.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.5	49.4	14.7	1	0	8.4
U2 - Efficacy additive alpha prior Beta(1, 1)	21.2	50.9	18.1	1.4	0.1	8.3
U3 - Efficacy additive alpha prior Beta(1, 1)	26	49.6	15.1	0.9	0	8.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	10.2	7	2.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	10.2	7.1	2.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	10.1	7.1	2.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.1	10.1	7.1	2.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	10	7.1	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	10	7.1	2.2	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.2	9.9	7.2	2.2	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.2	9.8	7.3	2.3	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.5	1.3	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.5	1.3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.5	1.3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.6	1.3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.6	1.3	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.5	1.3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.6	1.3	0.7	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.3	44.8	27.7	9.7	1.2	4.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.1	45.8	23.4	7.8	0.6	4.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	41.3	30	9.5	1.6	4.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.4	45.1	23.3	7.4	0.6	4.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.6	43.1	29.4	9.2	1.5	4.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.9	43.9	23.7	6.5	0.8	4.2
U2 - Efficacy additive alpha prior Beta(1, 1)	13	42.6	29.5	9.1	1.5	4.3
U3 - Efficacy additive alpha prior Beta(1, 1)	21.1	43.4	23.6	6.8	0.8	4.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.6	8.7	8.3	4.6	2.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.7	8.5	8.3	4.6	2.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	8.6	8.3	4.5	2.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	8.5	8.4	4.5	2.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	8.4	8.4	4.6	2.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.7	8.3	8.4	4.6	2.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.8	8.4	8.3	4.6	2.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.7	8.4	8.3	4.7	2.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.7	1.8	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.8	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.7	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.8	1.6	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.8	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.8	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.6	1.8	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.8	1.6	0.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	59.9	1.6	0	0	0	38.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	60.5	1.4	0	0	0	38.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	60.2	1.7	0	0	0	38.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	60.2	1.5	0	0	0	38.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	59.8	1.6	0	0	0	38.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	59.9	1.4	0	0	0	38.7
U2 - Efficacy additive alpha prior Beta(1, 1)	59.7	1.5	0	0	0	38.8
U3 - Efficacy additive alpha prior Beta(1, 1)	60.5	1.2	0	0	0	38.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	2.1	0.6	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	2	0.6	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.9	2.1	0.6	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.9	2.1	0.6	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.9	2	0.6	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.9	2	0.6	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20.9	2	0.6	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.9	2	0.6	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.7	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.3	55.1	12.4	1.9	0	4.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.5	54.7	10	1.5	0	4.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.1	53.8	13.2	1.6	0	4.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.1	52.4	9.6	1.6	0	4.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.6	54.2	13.2	1.7	0	4.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33	51.1	9.8	1.6	0	4.5
U2 - Efficacy additive alpha prior Beta(1, 1)	26.5	54.1	13.4	1.5	0	4.5
U3 - Efficacy additive alpha prior Beta(1, 1)	32.9	50.9	10.1	1.8	0	4.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	12.1	6.5	1.7	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	12.1	6.6	1.7	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	12	6.5	1.7	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	12	6.5	1.7	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	12	6.6	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	12	6.5	1.7	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.9	11.9	6.6	1.7	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.9	11.9	6.6	1.7	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.1	1.3	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.3	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.1	1.3	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4	1.3	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	1.3	0.5	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.6	5.8	0.2	0	0	74.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.2	5.7	0	0	0	74.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.8	5.5	0.1	0	0	74.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.3	5	0	0	0	73.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.5	5.8	0.1	0	0	73.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.9	5	0.1	0	0	74
U2 - Efficacy additive alpha prior Beta(1, 1)	20.2	6.1	0.2	0	0	73.5
U3 - Efficacy additive alpha prior Beta(1, 1)	21.2	4.5	0.1	0	0	74.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	2.7	0.6	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	2.7	0.6	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.2	2.7	0.6	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.2	2.7	0.6	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.2	2.7	0.6	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.3	2.6	0.6	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.2	2.7	0.6	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.2	2.7	0.6	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.5	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	1.9	4.8	43.8	48.5	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.4	2.4	6.5	47.4	41.2	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	1.6	4.2	43.9	49	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	2.7	2.8	6	45.3	43.1	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	1.2	4.5	42.9	50	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.9	2.4	6.2	44.1	44.3	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	1.4	4.7	42.8	49.9	0
U3 - Efficacy additive alpha prior Beta(1, 1)	2.9	2.4	6.1	43.4	45.1	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.3	3.8	8.2	11.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.3	3.8	8.2	11.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.3	3.2	3.8	8	11.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.3	3.2	3.8	8	11.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.3	3.2	3.8	7.8	11.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.3	3.2	3.8	7.9	11.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.3	3.2	3.8	7.8	11.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.3	3.2	3.8	7.8	11.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.1	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.5	1.2	3	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.5	1.2	3	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.2	3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.2	3	1.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	34	28.3	32.3	4.7	0	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	38.7	28.9	27.9	3.8	0	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.9	26.3	33.1	5	0	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	41.5	26.5	27.4	3.9	0	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.1	26.6	33	4.5	0.1	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	42.8	25.1	27.3	4.1	0	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	36	27	32.5	3.7	0.1	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	43.1	25	27.4	3.8	0	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	7.8	9.4	2.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	7.8	9.4	2.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	7.9	9.3	2.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	7.7	9.3	2.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	7.7	9.4	2.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	7.7	9.3	2.8	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.5	7.6	9.5	2.7	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.5	7.7	9.4	2.8	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.1	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.1	1.5	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.1	1.5	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.1	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.1	1.5	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.1	1.5	0.9	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29	26.5	28.5	5.8	0.1	10.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.3	26.1	24.2	5.2	0	10.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.2	25.8	29.6	5.3	0.1	10
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.8	25.5	24.6	5.9	0.1	10.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.4	25.9	28.4	6.1	0.1	10.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.5	25.5	24.5	5.4	0.1	10
U2 - Efficacy additive alpha prior Beta(1, 1)	30.5	25.5	28.4	5.3	0.1	10.2
U3 - Efficacy additive alpha prior Beta(1, 1)	36.8	23.1	24.5	5.3	0.1	10.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11	6.5	8	2.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11	6.5	7.9	2.9	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.9	6.4	8	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	6.4	8	3	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	6.3	8	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	6.4	8	3	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	11	6.4	8	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	11	6.4	8	3	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.9	1.3	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.3	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.3	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.3	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.9	1.3	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.9	1.3	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.9	1.2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.9	1.3	0.9	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.2	21.2	41.9	9.2	1	5.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.8	22.9	37.6	7.7	0.6	5.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.6	22.6	41.8	8.9	0.6	5.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.4	21.5	39.3	7.9	0.4	5.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20	21.5	43.3	8.5	1.2	5.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.5	21.5	38.5	7.4	0.6	5.5
U2 - Efficacy additive alpha prior Beta(1, 1)	20.3	22	42.6	8.7	0.9	5.5
U3 - Efficacy additive alpha prior Beta(1, 1)	26.4	20.8	39.5	7.1	0.7	5.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.3	6.3	9.4	4	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.2	6.4	9.4	4	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	6.2	9.4	4	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.3	6.2	9.4	4	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.3	6.2	9.4	4	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.3	6.1	9.6	3.9	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.3	6.2	9.4	4	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.3	6.1	9.4	4	1.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.6	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.9	1.6	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.9	1.6	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.6	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.9	1.6	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.9	1.6	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.9	1.5	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.9	1.5	1.1	0.3	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.6	37.6	6.8	0.3	0	33.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.8	38.8	5.7	0.3	0	33.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.1	37.2	6.7	0.4	0	33.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.3	38.1	5	0.3	0	33.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.9	37.5	6.8	0.4	0	33.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23	37.3	5.8	0.5	0	33.4
U2 - Efficacy additive alpha prior Beta(1, 1)	20.8	37.7	7.7	0.3	0	33.5
U3 - Efficacy additive alpha prior Beta(1, 1)	23.2	38.1	5.1	0.2	0	33.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12	8.4	4.3	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.1	8.4	4.3	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.1	8.3	4.3	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.1	8.4	4.2	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	8.2	4.4	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.2	8.3	4.3	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.2	8.3	4.3	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.2	8.3	4.2	0.9	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.7	0.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.7	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.7	0.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.7	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.7	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.7	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.7	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.7	0.8	0.3	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.7	23.3	4.1	0.3	0	2.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	72.4	21.5	3.2	0.3	0	2.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	70	22.9	4.1	0.3	0	2.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	73.5	20.3	3.1	0.2	0	2.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	71.3	21.7	3.9	0.4	0	2.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	75	19.1	3.1	0.2	0	2.6
U2 - Efficacy additive alpha prior Beta(1, 1)	71	21.7	4.2	0.4	0	2.7
U3 - Efficacy additive alpha prior Beta(1, 1)	74.5	19.7	3	0.2	0	2.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.2	7.1	3.6	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.3	7.1	3.5	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.3	7.1	3.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.3	7	3.6	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.4	6.9	3.6	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.4	6.8	3.7	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.4	6.9	3.6	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.4	6.9	3.6	0.5	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.1	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.1	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2	0.7	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21	13.2	15.7	37	12.3	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.8	14.5	13.8	32.9	10.2	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.9	14.3	15	36.3	12.7	0.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.1	14.4	12.4	33.6	9.7	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.8	15.3	17.1	35.6	12.4	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.7	15	12.7	33.2	9.6	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	19.1	14.7	16.3	36.2	12.9	0.8
U3 - Efficacy additive alpha prior Beta(1, 1)	28.7	13.9	13.9	32.2	10.5	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	4.5	5.7	8.1	6.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	4.5	5.7	8.1	6.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	4.4	5.7	8	6.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	4.3	5.7	8.1	6.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	4.4	5.8	8	6.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	4.4	5.7	8.1	6.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.2	4.4	5.8	8	6.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.2	4.4	5.7	8	6.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.7	1.5	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	1.5	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.7	1.6	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.7	1.5	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	1.5	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	1.5	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	1.6	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	1.5	2.2	0.9	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.6	35	25.6	25.2	6.5	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	41.9	24.3	19.5	3.9	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.5	32.8	27.3	26.3	6	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	41.9	24.2	19.3	4.5	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.8	32.2	29.2	23.8	6.9	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.4	37.8	27.3	19.2	4.2	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	8.3	32.2	29.4	23	7	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	11.2	37.1	27.4	19.7	4.5	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.7	6.4	7.7	6.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.7	6.4	7.6	6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	5.6	6.5	7.7	6.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	5.6	6.4	7.6	6.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	5.6	6.5	7.6	6.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	5.5	6.5	7.6	6.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.2	5.5	6.6	7.5	6.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.2	5.5	6.5	7.6	6.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.7	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.7	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.4	1.7	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	1.7	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.7	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.7	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	1.7	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	1.7	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	13.2	35	27.3	14.4	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.4	16.5	34	24.1	9.4	0.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	12.4	35.4	28	14.2	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.9	16.1	36	22.8	8.6	0.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	11.9	35.9	25.2	15.3	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.8	16.8	35	20.9	8.9	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	10.5	12.4	34.9	26	15.6	0.6
U3 - Efficacy additive alpha prior Beta(1, 1)	18	16.4	35	21.4	8.6	0.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	4.3	7.6	7	7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	4.3	7.6	7	7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	4.2	7.6	7.1	7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	4.2	7.6	7	7.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	4.2	7.6	7	7.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	4.2	7.6	7	7.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4	4.2	7.5	7	7.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4	4.2	7.5	7	7.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2	2.6	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2.1	2.6	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	2	2.6	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	2	2.5	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	2	2.5	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	2	2.5	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	2	2.5	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	2	2.5	1.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.3	5.6	12.7	40	34.3	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.6	6.6	12.4	40.6	27.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.3	5.8	12.8	39.1	34.9	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.5	6.5	12.7	38.3	29.9	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.7	5.8	13.5	38.9	35	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.1	7.2	13.5	35.8	30.3	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	6.5	6.2	12.9	38.7	35.6	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	12.6	7	14	36.8	29.5	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.6	5.2	8	9.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.7	5.2	8	9.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.6	5.2	7.9	9.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.6	5.2	7.8	9.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.6	5.2	7.8	9.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.6	5.2	7.8	10	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.6	5.3	7.6	10	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.6	5.2	7.7	10	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.8	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.8	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.7	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.7	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1.4	2.7	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1.4	2.7	0.9	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.4	12.2	47.8	18.4	6.2	2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.7	13.9	44.9	16.4	3.2	1.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.3	14.4	47.2	18.9	6.2	2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.3	14.4	43.3	15.3	3.8	1.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.3	14.1	47.1	17.8	5.7	2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22	13.6	42.6	15.7	4.1	2
U2 - Efficacy additive alpha prior Beta(1, 1)	11.6	14	47.2	19.2	6	2
U3 - Efficacy additive alpha prior Beta(1, 1)	21.5	13.7	43.5	15	4.3	2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.3	5.6	10	5.3	3.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	5.7	9.9	5.3	3.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	5.6	10	5.3	3.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	5.6	9.9	5.4	3.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	5.6	10	5.3	3.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	5.6	10	5.3	3.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.3	5.6	10	5.3	3.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.3	5.6	10.1	5.2	3.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.3	1.9	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.3	2	1.7	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.2	2	1.6	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.2	1.9	1.6	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.2	1.9	1.6	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.2	1.9	1.6	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.2	1.9	1.6	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.2	1.9	1.6	0.6	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.8	39.8	31.2	11.9	4.2	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	44.1	26.4	10.4	2.6	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12	39.5	31.9	11.9	4.6	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17	43.6	27.1	9.3	2.9	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12	38.6	32.8	11.5	5	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.5	41.5	27.5	10	3.4	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	13	36.3	33.3	12.3	5	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	17.7	39.8	27.8	11.1	3.5	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	8.2	8.9	5	3.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	8.3	8.9	5	3.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	8	9	5	3.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	8	8.9	5.1	3.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	7.9	9.1	4.9	3.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	7.9	9.1	5.1	3.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.8	7.9	9	5.1	3.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.8	7.9	9	5	3.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.1	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.1	1.6	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.1	1.6	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.6	1.3	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.6	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.1	1.6	1.3	0.5	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.4	40.2	15.9	6.3	2.3	14.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.5	42.5	11.6	4.9	1.4	15.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.3	41	15.4	6.1	2.1	15.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.8	39.1	11.9	5.8	1.3	15.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.8	39.7	15.6	6.4	2.5	15
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.6	38	12	5.4	1.9	15.1
U2 - Efficacy additive alpha prior Beta(1, 1)	21.5	38.9	15.4	6.6	2.1	15.5
U3 - Efficacy additive alpha prior Beta(1, 1)	28.5	38	11.5	5.1	1.9	15
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	8.2	5.8	2.8	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	8.2	5.7	2.8	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	8.2	5.8	2.8	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	8.1	5.8	2.8	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	8.1	5.8	2.8	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	7.9	5.9	2.8	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.8	7.9	5.9	2.8	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.7	8.1	5.8	2.8	1.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.5	1.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1.2	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.5	1.2	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.5	1.2	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.5	1.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.2	0.9	0.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.2	49.1	14.1	3.1	0.4	7.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.5	47.7	10	2.4	0.3	7.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.3	50.9	13.6	2.9	0.4	6.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32	48.2	10.2	2.2	0.3	7.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.4	49.5	14.7	2.8	0.4	7.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.2	46.8	10.8	1.7	0.3	7.2
U2 - Efficacy additive alpha prior Beta(1, 1)	23.6	51	14.8	3.1	0.4	7.1
U3 - Efficacy additive alpha prior Beta(1, 1)	33.2	46.5	10.6	2.1	0.5	7.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	10.7	6.1	2.4	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.6	10.7	6.1	2.4	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.7	10.6	6.2	2.4	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.7	10.5	6.2	2.4	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	10.5	6.2	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	10.5	6.2	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.7	10.5	6.3	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.8	10.5	6.3	2.3	0.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.1	1.3	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.1	1.3	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.1	1.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.1	1.3	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.2	1.3	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.1	1.3	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.1	1.3	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.2	1.3	0.9	0.2	